ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-13 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

A machine adds 4 to an input, then squares the result. Explain why $(x + 4)^2$ matches but $x^2 + 4$ does not.

2. I can use input-output rules to represent relationships.

A function machine triples the input and subtracts 2. What output comes from input 5?

3. I can use input-output rules to represent relationships.

Evaluate.

$f(x) = 2x + 5;\ f(-3)$

4. I can use input-output rules to represent relationships.

Write a rule.

Inputs 0,1,2 give outputs 4,7,10.

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Identify the repeated input.

x	-1	2	-1
y	3	5	8

6. I can distinguish functions from relationships that are not functions.

Pairs $(1, 5), (2, 5), (3, 5)$ all have output 5. Is this a function? Explain.

7. I can distinguish functions from relationships that are not functions.

Does $\{(1, 2), (1, 3), (2, 4)\}$ represent a function? Explain.

8. I can distinguish functions from relationships that are not functions.

For $y = 2x - 5$, find the output when $x = -1$.

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

9. I can connect linear growth in tables, graphs, and equations.

Find the change per step for outputs 5, 9, 13, 17. Is $(1, 2), (2, 2), (1, 3)$ a function?

10. I can connect linear growth in tables, graphs, and equations.

Write an equation.

Slope 4; y-intercept −2.

11. I can connect linear growth in tables, graphs, and equations.

A table has $x = 0, 1, 2, 3$ and $y = 3, 7, 11, 15$. Write a rule.

12. I can connect linear growth in tables, graphs, and equations.

x	0	1	2	3
A	1	4	7	10
B	1	4	9	16

Which is linear? Give numerical evidence.

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

13. I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0, 2)$ and $(3, 8)$.

14. I can calculate and interpret slope as a constant rate of change.

Find the slope through (0,3) and (4,11). Then evaluate $f(2)$ for $f(x) = 3x - 1$.

15. I can calculate and interpret slope as a constant rate of change.

As x rises by 4, y falls by 12. Find the slope.

16. I can calculate and interpret slope as a constant rate of change.

Find the run.

$(-1,8)$ to $(6,10)$

I CAN COMPARE VERTICAL AND HORIZONTAL CHANGES TO FIND SLOPE.

17. I can compare vertical and horizontal changes to find slope.

Find the slope.

Rise −6, run 3

18. I can compare vertical and horizontal changes to find slope.

Calculate the slope.

(0,−5) to (4,3)

19. I can compare vertical and horizontal changes to find slope.

Find rise/run from $(-2,1)$ to $(2,-3)$. Then find y when $x = 4$ in $y = 2x + 1$.

20. I can compare vertical and horizontal changes to find slope.

Find Δy and Δx from $(1, 3)$ to $(4, 9)$.